

RGIPT

Subject: Artificial Intelligence

Class: 12

Time Allowed: 60 minutes

Maximum Marks: 10

All questions are compulsory unless stated otherwise.

Name: _____

Roll Number: _____

Class: 12 Section: _____

Section A: Multiple Choice Questions

Choose the correct answer for each question. Each question carries 1 mark.

1. [Easy] What is the main drawback of character-level tokenization? a) Requires many model parameters b) Extreme granularity requiring many processing steps c) Cannot handle spaces d) Leads to higher semantic similarity [1 Marks]
2. [Moderate] Which algorithm merges the most frequent adjacent token pairs to build a subword vocabulary? a) WordPiece b) Unigram LM c) Byte Pair Encoding (BPE) d) SentencePiece [1 Marks]
3. [Challenging] Why are sinusoidal positional encodings preferred in the original Transformer architecture? a) They are learned during training and improve accuracy b) They require no learned parameters and encode relative positions linearly c) They reduce the dimensionality of embeddings d) They encode language-specific grammar rules [1 Marks]
4. [Easy] What does a token embedding vector primarily represent? a) The token's frequency in the training corpus b) The token's position in the sentence c) The semantic meaning of the token in a high-dimensional space d) The token's character count [1 Marks]

Section B: Short Answer Questions

Answer the following questions briefly. Each question carries 2 marks.

1. [Moderate] Explain how the "unknown" problem arises in word-level models and how subword tokenization mitigates it. [2 Marks]
2. [Challenging] Discuss the impact of tokenization on model cost and context limits, using GPT-4's token limit as an example. [2 Marks]
3. [Moderate] Describe how positional encoding enables a transformer to differentiate the sentences "The dog bit the man" and "The man bit the dog". [2 Marks]

End of Question Paper

Answer Key:

1. b
2. c
3. b
4. c
5. Word-level models have a fixed vocabulary; any word not in that list becomes an out-of-vocabulary (OOV) token, losing its meaning. Subword tokenization breaks rare or unseen words into smaller known pieces, allowing the model to represent the word using familiar subword tokens instead of a single unknown token.
6. API billing and the context window are measured in tokens, not characters or words. Efficient tokenization (fewer tokens per word) lets more content fit within GPT-4's 128K token limit and reduces the monetary cost per request. Poor tokenization inflates token counts, shrinking usable context and increasing expenses.
7. Positional encodings add a unique vector to each token based on its index. Thus, the same words acquire different positional vectors in the two sentences, producing distinct combined embeddings. The attention mechanism can then attend to word order, resulting in different internal representations for the two sentences.